

A Physiologically Interpretable Multimodal Model for Joint Sleep Staging and Respiratory-Event Detection in Subacute Ischemic Stroke

MD WAHIDUZZAMAN SUVA¹, ESM-E MOULA CHOWDHURY ABHA¹, MD IMTIAJ ALAM SAJIN¹,
AND MD IFTEKHARUL MOBIN¹

¹Department of Computer Science and Engineering, American International University-Bangladesh, Dhaka 1229, Bangladesh

Corresponding author: Md Iftekharul Mobin (e-mail: iftekhar.mobin@aiub.edu).

The authors received no specific funding for this work.

ABSTRACT Sleep-disordered breathing is common in the weeks after an ischemic stroke and predicts worse recovery, yet automatic sleep analysis reads only the electroencephalogram and reports only sleep stages. A clinical polysomnogram records far more: eye movements, muscle tone, the electrocardiogram, airflow, breathing effort, and blood oxygen saturation. In this population those channels carry the disease itself, and the median patient in this cohort has a scored respiratory event in 13% of epochs. We present MM-Net, a two-stream multi-task model that reads the full polysomnogram and produces two clinical outputs from a single forward pass: the sleep stage of every 30-second epoch and a per-epoch respiratory-event label, with a direct cardiorespiratory path that delivers breathing effort and desaturation to the respiratory decision undiminished by the staging objective. On iSLEEPS, the first public polysomnography corpus of subacute ischemic stroke (99 patients, 92,560 epochs), under patient-independent ten-fold cross-validation, MM-Net stages sleep at 0.739 accuracy (Cohen's κ 0.634) and detects respiratory events at 0.782 AUC with 2.76 M trainable parameters. We benchmark ten deep architectures retrained on this cohort and three applied zero-shot from healthy sleep, all on identical folds; every retrained baseline lands between 0.613 and 0.686 accuracy. Three are reimplemented here, each recovering its published Sleep-EDF accuracy to within 0.03, which indicates that the ceiling belongs to the cohort and not to our implementations. Cross-validated on healthy sleep the same architecture reaches 0.856, losing 0.115 accuracy crossing into stroke against 0.140 to 0.148 for those baselines. Zero-shot on two independent public corpora, the respiratory read-out holds at 0.780 AUC against 0.782 in-domain, while staging rises on healthy sleep and falls 0.090 on a sleep-disorders cohort. Code, features and checkpoints are released.

INDEX TERMS Cardiorespiratory signals, feature fusion, ischemic stroke, multi-task learning, multimodal learning, polysomnography, sleep apnea, sleep staging.

I. INTRODUCTION

SLEEP is scored in 30-second epochs into five stages, Wake (W), N1, N2, N3, and rapid-eye-movement (REM), following the American Academy of Sleep Medicine (AASM) rules [1]. The overwhelming majority of automated methods read a single electroencephalogram (EEG) channel and ask one question: how accurately can the five stages be classified? Two decades of work on that question have moved from convolutional and recurrent encoders [2]–[4] through attention and sequence models [5]–[7] to large self-supervised EEG foundation models [8]–[11]. On healthy sleepers these

methods reach roughly 85% accuracy, and recent reviews argue the field is near a ceiling set by inter-scorer disagreement [12].

A clinical polysomnogram (PSG), however, is a recording of the whole sleeping body. It carries the EEG, eye movements, muscle tone, the electrocardiogram (ECG), nasal and oral airflow, thoracic and abdominal breathing effort, and peripheral oxygen saturation (SpO_2). These cardiorespiratory channels are where sleep-disordered breathing writes itself: a paused breath, a chest that keeps straining against a closed airway, a fall in blood oxygen, and the arousal that rescues

the breath. A model that reads only the EEG cannot see any of this, and so cannot report on the disorder that most damages a patient's sleep. We use the full polysomnogram, and we ask a question that the staging race does not: what does a model gain, and on which clinical task, when it reads the whole recording rather than the brain alone.

We study this on iSLEEPS [13], released in 2026 as the first public PSG corpus dedicated to subacute ischemic stroke. The cohort is well matched to the question. Sleep-disordered breathing is common in the weeks after a stroke and is tied to worse recovery [14], [15], and in this corpus the median patient carries a scored respiratory event in 13% of epochs, with 77 of 96 patients above 5% (Fig. 1). The cardiorespiratory signal is therefore abundant and clinically meaningful in exactly the population where breathing matters most. The published staging baselines on this corpus reach 74.7% accuracy for an LSTM, about ten points below the healthy state of the art, and a concurrent study reports that deep networks attend to non-physiological regions on this cohort [16], so the setting also stresses every staging architecture we can bring to it.

We present a two-stream, multi-task model that treats staging and respiratory-event detection as one joint problem, and we place it inside a broad benchmark on the same folds. Our contributions are:

- 1) We introduce MM-Net, a two-stream multi-task model that stages sleep at 0.739 accuracy (κ 0.634) and, from the same forward pass, detects respiratory events at 0.782 AUC, with a direct cardiorespiratory path to the respiratory head so cardiorespiratory cues reach the decision undiminished by a fusion trained mostly for staging; removing that path costs 0.006 respiratory AUC and leaves staging unchanged (Section V). The ablation identifies respiratory effort, not oxygen saturation, as the channel the read-out depends on (Section VI-F).
- 2) We benchmark the model against thirteen baseline configurations on identical patient-independent folds, covering convolutional, recurrent, attentional and fully convolutional families, and reimplement three published architectures (U-Time, TinySleepNet, SleepTransformer) so that every comparison runs through one pipeline (Table 4).
- 3) We establish that the accuracy ceiling on this cohort belongs to the patients and not to our implementations. Each reimplementation recovers its published Sleep-EDF accuracy to within 0.03, every retrained baseline lands between 0.613 and 0.686 here, and the proposed model loses 0.115 accuracy crossing from healthy sleep to stroke against 0.140 to 0.148 for those baselines (Section VI-L).
- 4) We show the respiratory read-out is physiologically grounded: removing the airflow channel from which the events were scored leaves detection at 0.778 AUC, because the decision rests on respiratory effort above all, together with oxygen desaturation, cardiac variability and cortical arousals, encoded redundantly enough that no single channel is indispensable (Section VI-G).
- 5) We attribute each output causally to the physiology that produces it. A leave-one-out modality-ablation grid shows removing the cardiorespiratory stream lowers only respiratory detection (-0.128 AUC, Wilcoxon on thirty fold-values, Holm $p < 10^{-7}$), while removing the ocular channel lowers only staging, giving a falsifiable account of what each modality contributes (Table 6).

Together, these results reframe the problem on this cohort: the useful gains come from reading the whole sleeping body and attributing each output to the physiology that produces it; deeper single-channel staging does not help on this cohort. The remainder of the paper develops the model (Section V) and tests each claim in turn.

II. RELATED WORK

A. SINGLE-CHANNEL DEEP SLEEP STAGING

Automated staging matured on one EEG channel. DeepSleepNet paired convolutional feature extraction with a bidirectional LSTM [2]; SeqSleepNet framed the night as sequence-to-sequence labeling [3]; AttnSleep introduced attention over multi-resolution convolutions [5]; TinySleepNet and U-Time reduced the parameter budget while holding accuracy [4], [7]; and XSleepNet fused raw and spectral views [6]. More recent backbones push this further: SleepTransformer adds interpretability and uncertainty to an attention model [17], L-SeqSleepNet models whole-night cycles [18], ZleepAnlystNet and FlexSleepTransformer explore separated training and flexible channel sets [19], [20], and linear-time state-space backbones reach competitive accuracy [21], [22]. These models set the accuracy on healthy corpora but depend on the cortical micro-events that stroke disturbs, which is why they transfer poorly to patients [16].

B. MULTI-CHANNEL AND FOUNDATION MODELS

Later work reads several PSG channels. U-Sleep stages from arbitrary EEG and EOG derivations across many cohorts [23], and self-supervised EEG foundation models such as LaBraM, CBraMod, EEGPT, and BIOT learn transferable representations from large unlabeled corpora [8]–[11]. SleepFM extends this idea to several PSG modalities [24], while cross-scale foundation models and standardized benchmarks continue to appear [25]–[27], self-supervised pretraining is now systematically evaluated for label efficiency [28], transfer and domain adaptation improve robustness across cohorts [29], and recent surveys and clinical benchmarks catalogue the field [30], [31]. The eye and muscle channels help most with REM and Wake, which a single EEG channel resolves poorly. Reported gains from adding respiratory channels to *staging* are small and cohort-dependent; our benchmark quantifies this directly on a high-burden clinical cohort and separates the staging effect from the respiratory one.

C. DETECTION OF SLEEP-DISORDERED BREATHING

A parallel line of work detects apnea and hypopnea from breathing signals. Airflow and effort shape, oximetry, and ECG-derived heart-rate variability are all informative, and oxygen desaturation is a defining marker of an event [32]. These detectors usually run at fine time resolution on the raw breathing waveforms and are trained on that task alone. We instead detect respiratory events on the staging epoch grid, jointly with staging, from a shared representation, so that one model produces both clinical outputs and the two tasks support each other during training.

D. SLEEP AND BREATHING AFTER STROKE

Sleep is both disturbed by stroke and predictive of recovery, and sleep-disordered breathing is common in the subacute phase and linked to poorer functional outcome [14], [15], [33]. Focal injury also changes the micro-events that scoring relies on, the spindles, slow waves, and muscle atonia [34]–[38]; quantitative EEG markers such as the brain symmetry index further track hemispheric injury [39]. A model that reports on breathing as well as stage is therefore of direct clinical value in this population. Sleep staging on ischemic stroke itself is nascent, reported only as baselines on the iSLEEPS corpus [13], and no prior method addresses staging and respiratory detection jointly, from the full polysomnogram, on a stroke population; we make this comparison quantitative in Table 12.

III. CRITICAL GAPS AND LIMITATIONS OF PRIOR WORK

Three gaps run through the literature above, and each maps onto one of our contributions. (i) *Single-task, single-signal design*. Almost every staging model reads one or a few EEG channels and predicts only the stage, while apnea detectors read breathing signals and predict only events; the two literatures rarely meet in one model, leaving the clinician with two separate tools. Contribution 1 answers this with a single model that reads the full polysomnogram and produces both outputs. (ii) *Evaluation on healthy sleepers*. Staging accuracy is reported almost entirely on healthy corpora such as Sleep-EDF [40], [41], where deep and foundation models reach a ceiling, while performance on focal brain injury is largely unmeasured. Contributions 2 and 3 benchmark strong deep architectures, and classical gradient-boosting pipelines [42]–[44] on a subacute-stroke cohort, show that the healthy-built networks collapse, and establish by reimplementing that the collapse is a property of the cohort. (iii) *Attribution, reliability, and cost*. Interpretability is usually post-hoc and unfalsifiable, predictive uncertainty is seldom modeled [45], and the compute and energy footprint of heavy models is rarely reported [46]. Contributions 4 and 5 answer this with a physiologically grounded, non-circular respiratory read-out and a causal modality-ablation that attributes each output to the channel its physiology governs.

TABLE 1. Stage distribution and respiratory-event prevalence in the corpus we model ($N = 99$).

Stage	W	N1	N2	N3	R
Share (%)	26.6	10.2	42.3	8.9	12.1

Respiratory events: 16.0% of epochs positive; median 13% per patient (range 0 to 60.5%); 77 of 96 patients above 5%.

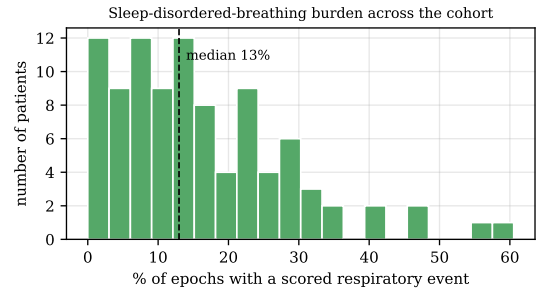


FIGURE 1. Sleep-disordered-breathing burden across the cohort. Each bar counts patients by the fraction of epochs that carry a scored respiratory event. The median patient sits at 13% and many exceed 30%, so the cardiorespiratory signal is abundant.

IV. DATASET AND PREPROCESSING

iSLEEPS [13] contains overnight PSG from 100 subacute ischemic-stroke patients (mean age 50.5; 77 male, 23 female), scored by experts into the five AASM stages. We exclude one recording after an audit: the data records of SN28 are byte-identical to those of SN15, a duplication consistent with the anonymization and re-numbering the dataset authors describe, and keeping it would place a training subject into the test fold in nine of ten fold assignments, giving $N = 99$. Sleep staging uses all 99 patients; the respiratory task uses the $N = 96$ with complete cardiorespiratory channels, and we state which N applies to each reported number. The corpus totals 92,560 epochs. The stage distribution (Table 1) is strongly imbalanced, with N2 dominant and N1 and N3 scarce, which is the clinical norm.

a: Channels we read, and channels we do not

A raw recording holds 22 traces. We read the 14 that carry sleep-stage or breathing information. The *neural stream* takes the four EEG derivations (C4:M1, C3:M2, O2:M1, O1:M2), the two EOG channels (E1:M2, E2:M2), and the submental EMG. The *cardiorespiratory stream* takes the ECG, nasal and oral airflow, thoracic effort, abdominal effort, the summed effort trace, SpO₂, and pulse. We leave out eight traces for stated reasons: periodic limb movement records a different disorder; snore is scored from the same airway sound as the respiratory events and would make the second task circular; plethysmography is already summarized by the SpO₂ and pulse channels we keep; body position is coarse and slow; and the ambient-light and battery traces are device housekeeping, not physiology. All neural channels are resampled to 100 Hz and the cardiorespiratory channels to a common 25 Hz grid; SpO₂, recorded natively at 4 Hz, is upsampled to this grid,

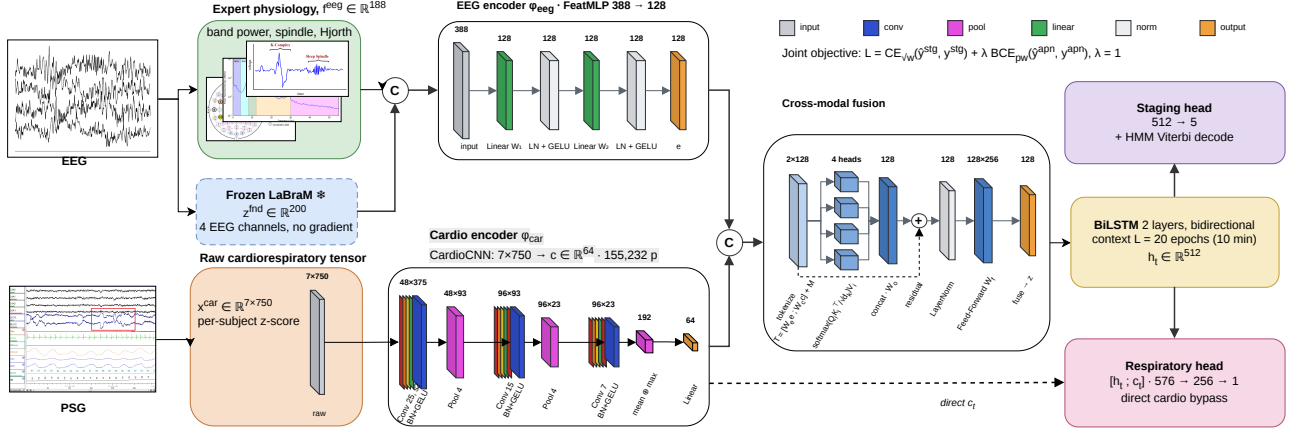


FIGURE 2. The two-stream, multi-task model. Layer shapes above each block, operation below; block height encodes units for the multilayer perceptron and temporal length for the convolutional stack. Dashed outlines mark frozen parameters. The junction marked C is the concatenation used for every reported result; the panel at lower right details the attention fusion evaluated against it in Section VI-F, which it matches to within noise.

so its effective resolution remains that of the 4 Hz oximeter. Signals are read in physical units, which matters because SpO_2 near 90%, pulse near 78 bpm, and microvolt EEG live on very different scales. Per-epoch respiratory labels are the corpus's scored events mapped onto the 30 s grid, and an epoch is positive when it overlaps an event.

V. PROPOSED METHOD

A. PROBLEM FORMULATION

A night is a sequence of T epochs. For epoch t we form a neural feature vector $\mathbf{f}_t^{\text{eeg}} \in \mathbb{R}^{388}$, formed by concatenating the 188 engineered descriptors below with a 200-dimensional embedding from a frozen pretrained EEG encoder (Section VI-M), and a raw cardiorespiratory tensor $\mathbf{x}_t^{\text{car}} \in \mathbb{R}^{7 \times 750}$, the seven channels at 25 Hz. The 14 engineered cardiorespiratory descriptors defined below are used by the baseline configuration against which the learned encoder is measured. We learn a subject-independent map

$$f_{\theta} : (\mathbf{f}_{1:T}^{\text{eeg}}, \mathbf{x}_{1:T}^{\text{car}}) \mapsto (\hat{\mathbf{y}}_{1:T}^{\text{stg}}, \hat{\mathbf{y}}_{1:T}^{\text{apn}}), \quad (1)$$

where $\hat{\mathbf{y}}_t^{\text{stg}} \in \Delta^4$ is a distribution over the five stages and $\hat{\mathbf{y}}_t^{\text{apn}} \in [0, 1]$ is the probability that epoch t contains a respiratory event (Fig. 2).

B. PHYSIOLOGICAL FEATURE REPRESENTATIONS

a: Neural features (188)

Per channel we take absolute and relative band powers from the Welch spectrum $P(f)$ in delta (0.5–4), theta (4–8), alpha (8–12), sigma (12–16), and beta (16–30) Hz; the relative power in band b is

$$\text{RP}_b = \frac{\int_{f \in b} P(f) df}{\int_{0.5}^{30} P(f) df}. \quad (2)$$

We add spectral entropy $H = -\sum_i p_i \log p_i$ with p_i the normalized spectrum, the 95% spectral-edge frequency, the three Hjorth descriptors [47], and time-domain statistics. Sleep

spindles are detected on the sigma-band analytic envelope $e(t) = |\mathcal{H}\{x_{\sigma}\}(t)|$ from the Hilbert transform $\mathcal{H}$, with density counted above a per-epoch threshold; slow-wave amplitude, ocular-movement energy on the EOG, and tonic EMG level complete the set.

b: Cadiorespiratory features (14)

From the seven cardiorespiratory channels we compute SpO_2 mean, minimum, variability, and desaturation depth (median minus minimum); pulse mean and variability; ECG amplitude and line-length; airflow amplitude and line-length; thoracic, abdominal, and summed effort amplitudes; and thoraco-abdominal asynchrony, the loss of coordination between chest and abdomen that marks an obstructed breath,

$$\text{asyn}_t = |\text{corr}(x_t^{\text{tho}}, x_t^{\text{abd}})|. \quad (3)$$

Both feature sets are standardized per patient, which removes person-to-person offsets so the model generalizes across people.

c: The learned cardiorespiratory encoder

The evaluated model does not read the 14 descriptors. Its cardiorespiratory branch is a three-layer convolutional encoder over the raw 7×750 tensor, with wide kernels and aggressive downsampling because respiratory events are slow: at 25 Hz the first kernel of 25 samples spans one second, and a hypopnea unfolds over 10 to 30. Channel widths are 48, 96 and 96, each followed by batch normalisation and GELU, with max-pooling by four after the first two. Mean and max pooling over time are concatenated and projected to the same 64 dimensions the feature MLP produced, so the encoder is a drop-in replacement and every other part of the network is unchanged. Section VI-M reports what that substitution is worth, and it is the single largest architectural effect in this paper.

C. TWO-STREAM ENCODERS

The neural stream is a two-layer perceptron with LayerNorm; the cardiorespiratory stream is the convolutional encoder described above, which emits a vector of the same width the perceptron it replaced produced,

$$e_t = \text{GELU}(\text{LN}(W_2 \text{GELU}(\text{LN}(W_1 f_t^{\text{eeg}})))) \in \mathbb{R}^{128}, \quad (4)$$

$$c_t = \text{CardioCNN}(x_t^{\text{car}}) \in \mathbb{R}^{64}. \quad (5)$$

LayerNorm rather than BatchNorm is used because the recurrent windows are short and mix patients within a batch, and the cardiorespiratory stream is deliberately narrower because the branch compresses a 7×750 tensor into the same 64 dimensions the feature encoder it replaced produced, which keeps the substitution the only variable.

D. FUSION

The two embeddings are concatenated and projected back to the stream width,

$$z_t = \text{GELU}(W_f [e_t; c_t]) \in \mathbb{R}^D, \quad D = 128. \quad (6)$$

This is deliberately the simplest fusion available, and it is what every number in this paper was produced with. We arrived at it by evaluating the obvious alternative first: the two embeddings as tokens carrying learned modality-type embeddings, refined by four-head scaled dot-product attention so each modality can read the other, then projected. That design matches concatenation to within noise (Section VI-F). Since the elaborate fusion buys nothing measurable here, we report and release the simple one, and the gain from the second stream should be read as coming from the added modality and not from how it is combined.

E. TEMPORAL DECODER AND TWO HEADS

Sleep is sequential and respiratory events cluster, so the fused vectors of an $L = 20$ epoch window (10 minutes) are read by a two-layer bidirectional LSTM. Each direction runs the standard recurrence

$$\begin{aligned} i_t &= \sigma(W_i[z_t, h_{t-1}]), & f_t &= \sigma(W_f[z_t, h_{t-1}]), \\ o_t &= \sigma(W_o[z_t, h_{t-1}]), & g_t &= \tanh(W_g[z_t, h_{t-1}]), \\ s_t &= f_t \odot s_{t-1} + i_t \odot g_t, & h_t &= o_t \odot \tanh(s_t), \end{aligned} \quad (7)$$

and the forward and backward states of the 256-unit layers are concatenated into $h_t \in \mathbb{R}^{512}$. The staging head is a linear map $\hat{y}_t^{\text{stg}} = \text{softmax}(W_s h_t)$. The respiratory head reads h_t together with a direct copy of the per-epoch cardiorespiratory embedding c_t ,

$$\hat{y}_t^{\text{apn}} = \sigma(W_2 \text{GELU}(W_1 [h_t; c_t])). \quad (8)$$

The connection exists so that the respiratory decision does not have to survive a fusion trained mostly for staging, which would weaken the mechanical signature of an obstructed breath along the way. Section VI-F identifies the channel that signature lives in, and it is respiratory effort, not the oxygen fall, that names the event. We isolate the connection by removing it and holding everything else fixed, on the

same thirty fold-values as every other figure here. Respiratory AUC falls from 0.782 to 0.776 (Wilcoxon $p = 0.043$) and average precision from 0.419 to 0.411 ($p = 0.42$), while staging is untouched at 0.739 accuracy ($p = 0.97$) and κ 0.634 ($p = 0.94$). The gain is small, about a sixth of a fold standard deviation, and the p -value is marginal enough that we would not rest an argument on it alone. What makes it worth reporting is where it lands rather than how large it is: the connection feeds the respiratory head and nothing else, and the respiratory head is the only output that moves.

The trained model holds 2.76 million parameters. The pretrained encoder that supplies the 200-dimensional neural embedding is run once offline and is not trained here, so it adds no trainable parameters to this count.

F. TRAINING AND DECODING

The two tasks are trained together with equal weight, $\mathcal{L} = \mathcal{L}^{\text{stg}} + \lambda \mathcal{L}^{\text{apn}}$ with $\lambda = 1$, combining a class-weighted staging cross-entropy and a positive-weighted respiratory cross-entropy,

$$\mathcal{L}^{\text{stg}} = - \sum_t \sum_k w_k y_{t,k}^{\text{stg}} \log \hat{y}_{t,k}^{\text{stg}}, \quad (9)$$

$$\mathcal{L}^{\text{apn}} = - \sum_t [\pi a_t \log \hat{y}_t^{\text{apn}} + (1 - a_t) \log(1 - \hat{y}_t^{\text{apn}})]. \quad (10)$$

The staging weights use the square root of inverse class frequency, $w_k \propto \sqrt{N/(Kn_k)}$ for $K = 5$ classes with n_k examples, which corrects the scarce N1 and N3 stages without giving up overall accuracy, and the respiratory term is positive-weighted by $\pi = n_-/n_+$ to match the 16% event rate. At inference the staging posteriors are smoothed by a hidden-Markov Viterbi pass whose transition matrix A^H and prior are counted from the training labels [48],

$$\delta_t(j) = \max_i [\delta_{t-1}(i) + \log A_{ij}^H] + \log \hat{y}_t^{\text{stg}}(j), \quad (11)$$

recovered by back-tracking, which restores the natural persistence of sleep. The respiratory head is left unsmoothed. Optimization uses AdamW with cosine annealing and early stopping on a held-out set of patients.

VI. EXPERIMENTS

A. PROTOCOL AND METRICS

All results use ten-fold, patient-independent cross-validation, so no patient appears in both training and test, and within each training split a disjoint set of patients is held out for early stopping. Staging is scored by accuracy, macro-F1 $\frac{1}{K} \sum_k \text{F1}_k$, and Cohen's κ . The respiratory task is scored by the area under the ROC curve (AUC) and average precision (AP), which suits the 16% event rate.

Every number we report is a mean over the ten folds with its standard deviation, and the folds are fixed across all configurations so that comparisons are paired. The proposed model, its ablations and the three reimplemented literature models all use the full ten; the earlier retrained baselines were

TABLE 2. Training configuration, shared across all folds, re-implemented baselines, and ablations.

Setting	Value
Framework / hardware	PyTorch, NVIDIA RTX 2060 (6 GB)
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
LR schedule	cosine annealing
Sequence length L	20 epochs (10 min)
Batch size	32
Max epochs / patience	45 / 8
Gradient clipping	5.0
Staging class weights	$\sqrt{\text{inverse-frequency}}$
Respiratory pos. weight	$\pi = n_- / n_+$
Loss balance λ	1.0
Staging post-processing	HMM Viterbi smoothing
Cross-validation	10-fold, patient-independent
Random seeds	42, 1, 7

run at five, which Table 4 states per row. Because a single training seed can flatter a result, we retrained the headline configuration under five seeds on the same folds. Staging accuracy varies by 0.005 across seeds against 0.020 across folds, and respiratory AUC by 0.001 against 0.038: fold-to-fold spread is four times the seed spread for staging and thirty times it for respiratory detection. The uncertainty in these results is therefore dominated by which patients fall in which fold, which is a property of a cohort this size, and not by the optimiser's starting point. Every headline figure in this paper is the mean over all thirty fold-values, ten folds by three seeds, and not a single run. The three seed means for respiratory AUC are 0.781, 0.783 and 0.781, so 0.782 is stable to the third decimal under reseeding.

B. IMPLEMENTATION AND HYPERPARAMETERS

The model is implemented in PyTorch and trained on a single NVIDIA RTX 2060 GPU (6GB) under Windows. Every run is seeded, so each is individually reproducible; headline figures average three seeds (42, 1, 7) over the ten folds, giving the thirty fold-values used throughout. Where a result is derived from stored predictions rather than retrained, it comes from the seed-42 run, and the text says so at that point. Table 2 lists the training configuration, which is identical across all folds, all baselines we re-implement, and every ablation, so that reported differences come from the model and not from tuning.

C. COMPARED METHODS

We compare the model against deep networks evaluated on the same folds: convolutional and recurrent networks trained from scratch (DeepSleepNet, AttnSleep, a four-channel CNN-BiLSTM), three further published architectures we reimplemented for this comparison (U-Time, a fully convolutional encoder-decoder; TinySleepNet, a compact convolutional-recurrent model; and SleepTransformer, a purely attentional one), a network transferred from Sleep-

TABLE 3. Reimplementation check for the models marked [†] in Table 4: ten-fold subject-independent cross-validation on the Sleep-EDF Cassette subjects 0–20, 40 recordings, single Fpz-Cz derivation. This is one subject more than the conventional Sleep-EDF-20 split, so the published accuracies opposite are not computed on an identical set. $\approx$ marks an accuracy inferred from the macro-F1 its paper reports.

Model	Published	Ours	Δ
U-Time [7]	≈ 0.79	0.791 ± 0.062	+0.001
TinySleepNet [4]	0.854	0.824 ± 0.045	−0.030
SleepTransformer [17]	0.848	0.823 ± 0.033	−0.025

EDF, and a raw-signal convolutional network on all 14 channels, alongside the published single-channel baselines for the corpus [13]. All numbers are patient-independent, ten-fold except for the earlier retrained baselines at five, at the hidden-Markov operating point where a model provides one.

D. STAGING BENCHMARK AND COMPARISON

Each reimplementation is checked against its published Sleep-EDF accuracy in Table 3; those published figures follow each paper's own protocol, which differs in subset and split, so they are consistency checks and not exact reproductions. Table 4 is organised in three blocks: architectures retrained on this cohort, architectures trained on healthy sleep and applied to it zero-shot, and the proposed model, with a final row cross-validating the proposed architecture on healthy sleep as a control on the cohort argument; its macro-F1 and κ are healthy-sleep figures and so are left out of the stroke columns, and quoted in the text instead. Dispersion is reported wherever per-fold values survive: five folds for the earlier retrained baselines, ten for the three reimplemented literature models, and thirty fold-values (ten folds, three seeds) for the proposed rows. The LSTM and CNN-ResNet18 rows are quoted from the corpus paper rather than re-run here, so their protocol is that paper's and not ours, and AttnSleep retained only aggregates, so neither carries an interval, and the zero-shot block is a single pass over all 99 patients with no fold structure. Sleep-EDF carries no cardiorespiratory channels, so every healthy-sleep figure exercises the neural branch alone and the comparison row there is MM-Net (neural only).

Table 4 shows a clear pattern: every deep network, whether trained from scratch, transferred from healthy sleep, or applied to the raw multichannel signal, stays between 0.61 and 0.69 accuracy, 15 to 20 points below the same architectures' published accuracy on healthy sleepers. The feature-based model reaches 0.739, above every deep architecture we retrained here, which matches the concurrent finding that depth alone does not help on this cohort [16]: with under a hundred patients a convolutional network cannot rediscover what decades of sleep science already encode in the features. It does not lead the corpus. The published single-channel LSTM reaches 0.747, above our 0.739, and we list it in Table 4 instead of leaving the best published number off the comparison. No method we are aware of exceeds 0.75 on this corpus. Section VI-K shows why: the staging learning curve

TABLE 4. Sleep staging on iSLEEPS, patient-independent, against every architecture we evaluated. [†] marks a model reimplemented by us from its paper (Table 3). “feat” marks engineered features. Cells are mean $\pm$ SD across folds where per-fold values were retained. Best per column in bold.

	Model	Input	Params $\downarrow$	Healthy	iSLEEPS (stroke)		
				Acc	Acc $\uparrow$	Macro-F1 $\uparrow$	κ $\uparrow$
<i>Deep networks, retrained on iSLEEPS</i>	LSTM [13]	1 EEG	n/r	—	0.747	—	—
	CNN-ResNet18 [13]	1 EEG	~ 11 M	—	0.617	0.544	0.480
	DeepSleepNet [2]	1 EEG	~ 21 M	—	0.615 ± 0.026	0.567 ± 0.031	0.483 ± 0.038
	U-Time [7] [†]	1 EEG	1.07 M	0.791 ± 0.062	0.646 ± 0.064	0.524 ± 0.057	0.506 ± 0.082
	AttnSleep [5]	1 EEG	~ 0.5 M	—	0.686	0.629	0.577
	TinySleepNet [4] [†]	1 EEG	1.45 M	0.824 ± 0.045	0.676 ± 0.050	0.587 ± 0.034	0.548 ± 0.062
	SleepTransformer [17] [†]	1 EEG	2.79 M	0.823 ± 0.033	0.683 ± 0.026	0.602 ± 0.030	0.555 ± 0.032
	CNN + BiLSTM	4 EEG	~ 0.6 M	—	0.613 ± 0.040	0.565 ± 0.034	0.485 ± 0.048
	Sleep-EDF transfer	4 EEG	~ 0.6 M	—	0.623 ± 0.020	0.582	0.488
<i>Trained on healthy sleep, applied zero-shot</i>	Raw multimodal CNN	14 ch	0.95 M	—	0.654 ± 0.033	0.610 ± 0.022	0.544 ± 0.037
	CNN	4 EEG	0.49 M	0.806	0.325	0.237	0.111
	CNN + BiLSTM (CRNN)	4 EEG	1.15 M	0.834	0.307	0.251	0.111
<i>Proposed feature-based multi-task model</i>	StagingSeqNet	4 EEG	1.08 M	0.868	0.308	0.232	0.093
	MM-Net (neural only)	7 ch feat	2.62 M	—	0.741 ± 0.017	0.668 ± 0.020	0.635 ± 0.022
<i>Same model, healthy sleep</i>	MM-Net (neural + cardio)	7 ch feat + raw	2.76 M	—	0.739 ± 0.020	0.670 ± 0.020	0.634 ± 0.024
	MM-Net (neural only)	2 EEG feat	2.62 M	0.856 ± 0.027	—	—	—

has flattened, so the ceiling is a property of this cohort rather than of any one model. We therefore do not position the model as a staging leader. What sets it apart is a second output: it is the only model that also detects respiratory events. A representative held-out night is shown in Figure 3: the predicted hypnogram follows the reference through the night’s cycles, and the stage-probability ribbon exposes where the model is confident and where it hesitates.

E. PER-CLASS STAGING PERFORMANCE

Table 5 breaks staging down by stage. The pattern is the one this cohort imposes on every method: N2, Wake, and REM are recovered well, N3 is moderate, and N1 is the hard stage, because it is transient and rare. The raw convolutional network reaches the highest N1 F1, which fits its sensitivity to short events, while the proposed model leads on the stable stages. The row-normalised confusion matrix (Fig. 4) shows where the errors land: N1 is absorbed mainly into Wake and N2, and part of N3 is read as N2. Comparing the neural-only and full rows isolates what the cardiorespiratory stream contributes to staging, and the answer is narrow: N1 improves by 0.019 (Wilcoxon $p = 0.045$ over thirty fold-values) and no other stage changes significantly ($p \geq 0.15$; REM shifts by 0.01 in Table 5, well inside that). That the gain lands on N1 is consistent with its physiology. It is the transitional stage, the hardest to score, and the one most often coincident with a respiratory event.

F. MODALITY-ABLATION GRID

This ablation is also our attribution method. Because the model is feature-based and never sees a raw spectrogram or a raw EEG trace, a gradient heat-map or Grad-CAM would have no faithful spatial support on the neural side, which is where a staging attribution would be read; we therefore attribute each output by *causal* intervention, deleting a physio-

TABLE 5. Per-class staging F1 (Wake, N1, N2, N3, REM), ten-fold patient-independent, at the hidden-Markov operating point. The neural-only and full rows are means over the same thirty fold-values. Best per column in bold.

Model	W $\uparrow$	N1 $\uparrow$	N2 $\uparrow$	N3 $\uparrow$	R $\uparrow$
Raw multimodal CNN	0.74	0.37	0.73	0.56	0.67
MM-Net (neural only)	0.79	0.28	0.80	0.71	0.76
MM-Net (neural + cardio)	0.79	0.30	0.80	0.71	0.75

logical channel and measuring the change. This is a falsifiable test of what each modality contributes, and not a post-hoc visualization. The proposed model stages sleep at 0.739 accuracy (κ 0.634) and detects respiratory events at 0.782 AUC (0.419 average precision). To test the paper’s central claim, that each modality carries the task its physiology governs, we remove each modality in turn and keep the rest, on the same folds, with three seeds giving thirty fold-values per row (Table 6). The separation between streams is clean, but within the cardiorespiratory stream the pattern is not the one the engineered features gave.

Removing the whole cardiorespiratory stream drops respiratory AUC from 0.782 to 0.654, the largest single effect in this study, and leaves staging untouched (0.741, n.s.). Removing the neural stream does the reverse: staging falls from 0.739 to 0.680 while respiratory detection holds. No neural ablation moves respiratory AUC and no cardiorespiratory ablation moves staging, so each stream serves the outcome its physiology governs.

Within the cardiorespiratory stream, however, only respiratory effort is individually necessary (-0.038 AUC, Holm $p < 0.001$). Removing SpO₂, pulse/HRV, ECG or airflow individually costs at most 0.005 AUC and none survives correction; SpO₂, which the engineered-feature configuration was built around, is here the most expendable channel

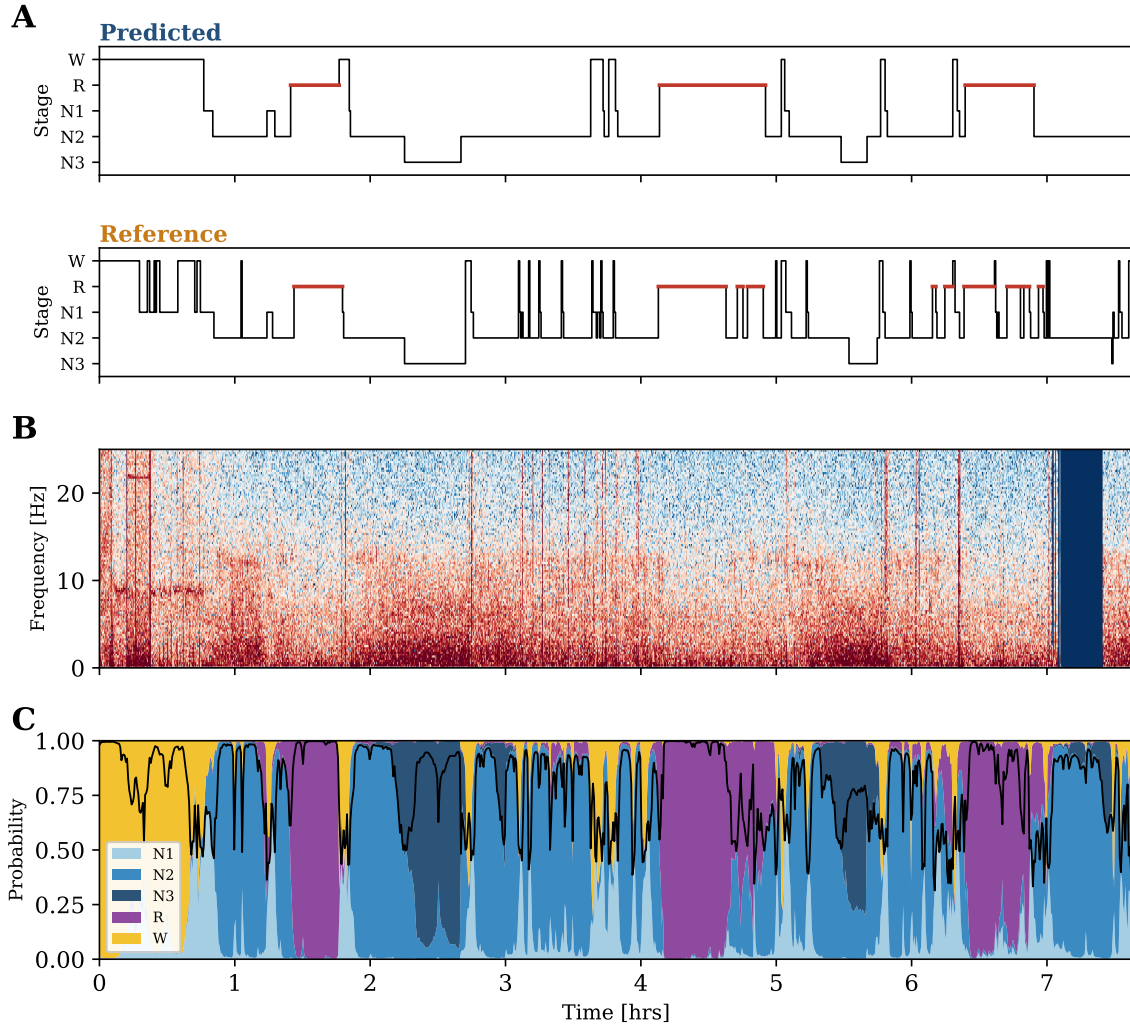


FIGURE 3. A representative held-out night (patient SN90, staging accuracy 0.80). (A) Predicted and reference hypnograms, REM highlighted. (B) EEG spectrogram, C4 derivation, 0–25 Hz. (C) Per-epoch stage-probability ribbon; the black line is the maximum posterior.

(-0.002 , $p = 0.61$). The five individual removals sum to -0.046 AUC against -0.128 for removing the stream as a whole, a factor of 2.8. The learned encoder therefore distributes respiratory information redundantly across channels: each is individually dispensable because the others overlap it, while the ensemble is essential. That redundancy is clinically favourable, not a weakness, since twelve of the hundred recordings are missing at least one cardiorespiratory channel, and it is physiologically coherent: the effort belts register the paradoxical thoraco-abdominal motion of an obstructive event as it happens, whereas desaturation is a lagged consequence of it. Attention and plain concatenation stage and detect alike, so the gain comes from the added modality, and the fusion mechanism has little effect.

G. PHYSIOLOGICAL VALIDITY OF THE RESPIRATORY READ-OUT

Because the events were scored from airflow, a model that keyed on the airflow channel would be right for a trivial

TABLE 6. Modality-ablation grid (leave-one-out), ten-fold patient-independent, three seeds. Each row removes one modality and keeps the rest; cells are mean $\pm$ SD over thirty fold-values. Largest effect per column in bold.

Modality removed	Staging acc $\uparrow$	Respiratory AUC $\uparrow$
none (full model)	0.739 ± 0.020	0.782 ± 0.038
EEG (neural)	0.680 ± 0.024	0.774 ± 0.041
EOG (ocular)	0.731 ± 0.023	0.777 ± 0.042
EMG (muscle)	0.735 ± 0.025	0.778 ± 0.041
SpO ₂	0.742 ± 0.019	0.780 ± 0.041
respiratory effort	0.741 ± 0.018	0.744 ± 0.049
pulse / HRV	0.741 ± 0.020	0.785 ± 0.035
ECG	0.737 ± 0.020	0.777 ± 0.043
airflow	0.740 ± 0.021	0.778 ± 0.039
all cardiorespiratory	0.741 ± 0.017	0.654 ± 0.041

reason. Removing the airflow features leaves detection essentially unchanged, at 0.778 AUC (Table 6), within one fold standard deviation of the full 0.782. The model therefore does not lean on the labeling signal. It reads above all the

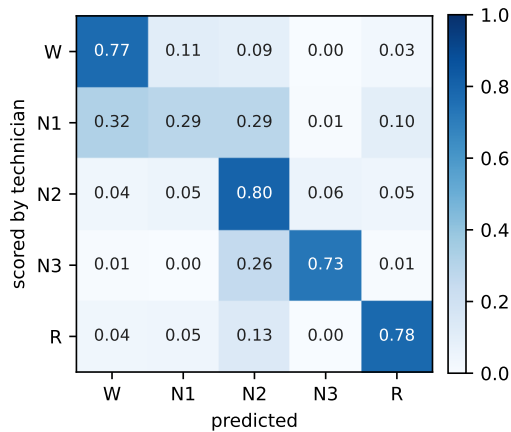


FIGURE 4. Row-normalised confusion matrix for the proposed model (pooled ten-fold test epochs, hidden-Markov operating point). Errors concentrate on N1, which is absorbed into Wake and N2.

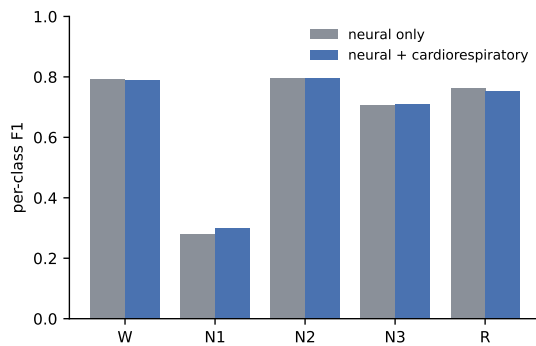


FIGURE 5. Per-class staging F1 for the neural-only and the neural-plus-cardiorespiratory model, both averaged over the same thirty fold-values.

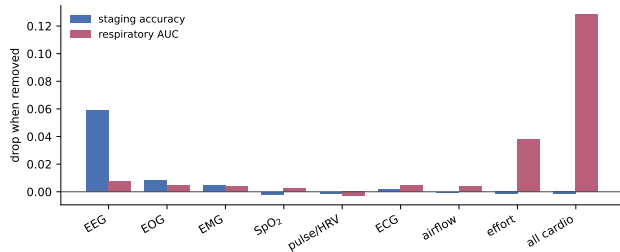


FIGURE 6. Modality-ablation grid. Drop in staging accuracy and in respiratory AUC when each modality is removed and the model retrained.

straining of the chest and abdomen (the only channel whose removal costs significantly), together with the fall in blood oxygen, the beat-to-beat change in the heart, and the cortical arousal that ends the event, none of which is individually indispensable because the learned encoder encodes them redundantly. This is the useful regime, because it detects the bodily consequences of disordered breathing, not a repeat of a measurement a technician has already made. It is also the physiologically expected one: the effort belts register the paradoxical thoraco-abdominal motion of an obstructive

event while it is happening, whereas desaturation follows it by several breaths.

H. RESPIRATORY BASELINES AND PER-EVENT-TYPE DETECTION

The respiratory task needs real competitors (Table 7), on the same folds. The classical baselines read the 14 engineered cardiorespiratory descriptors; the proposed model reads the raw signal through its learned encoder, and its own score on the 14 descriptors is 0.698 (Table 9), which is the like-for-like comparison. A clinical desaturation rule reaches 0.596 AUC, logistic regression 0.582, and gradient boosting 0.670; a random classifier scores 0.157 average precision, the event prevalence. The proposed model reaches 0.782 AUC and 0.419 average precision, above gradient boosting but by a modest margin, so the contribution is the joint single pass and the physiological attribution above; we do not claim state-of-the-art detection accuracy. Detection also separates by event type: the model scores 0.748 AUC on hypopneas (the mild, dominant 81% of events), 0.898 on obstructive apneas, and 0.935 on central apneas. It is strongest on the most severe events, and the overall AUC is held down by the abundant hypopneas.

Scoring at the event level instead of the epoch level changes the picture, and not favourably. Grouping consecutive flagged epochs into events within each patient, a decision threshold of 0.5 detects 55% of scored events at 2.6 false alarms per hour; lowering the threshold to 0.2 raises event sensitivity to 86% but flags 68% of all epochs. More tellingly, the model recovers the wrong *number* of events: a mean predicted rate of 4.5 events per hour against a scored 11.1, and a per-patient rank correlation of only $\rho = 0.142$ ($p = 0.17$, $n = 96$), far weaker than the $\rho = 0.491$ obtained from the aggregated epoch probability. Consecutive events separated by a few epochs merge into one flagged run, so an epoch-level detector systematically under-counts. This is why we report a burden and not an index, and why event-level scoring is the natural next step rather than a formality.

Figure 7 shows the operating characteristics behind these numbers, computed from the pooled held-out predictions of all ten folds. The precision–recall panel is the more informative of the two: respiratory events occupy 16% of epochs, and under that imbalance a ROC curve flatters any detector. Read at fixed sensitivity, the trade-off is explicit: at 0.80 sensitivity the model holds 0.47 specificity and 0.22 precision, which is a screening signal, not a scoring-grade detector, and we describe it as such. We also note that pooling every fold's epochs into a single curve is not the same estimator as averaging the ten per-fold scores: pooling weights folds by epoch count and mixes their score scales, giving a pooled AUC of 0.777 against the cross-validated 0.782 ± 0.038 reported in Table 7. Both are stated so the difference is visible, not hidden; at 0.005 it is an eighth of one fold standard deviation.

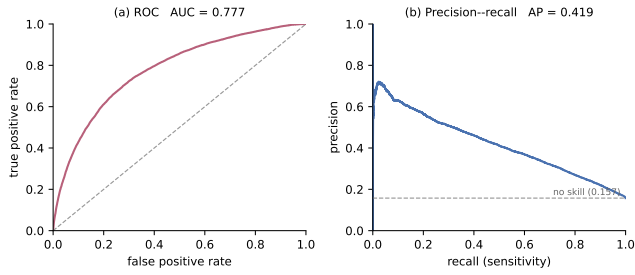


FIGURE 7. Respiratory-event detection, operating characteristics, pooled over the ten patient-independent test folds. (a) ROC; the dashed line is chance. (b) Precision-recall; the dashed line is the 0.157 event prevalence, the no-skill floor.

TABLE 7. Respiratory-event detection, ten-fold patient-independent. (a) Detectors trained on the same folds. The classical baselines read the 14 cardiorespiratory descriptors; MM-Net reads the raw signal through its learned encoder. Chance-level average precision equals the event prevalence, 0.157. (b) AUC by event type over the 96 patients with event-type annotation. Best per column in bold.

(a) Detection method		
Method	AUC↑	AP↑
Desaturation rule	0.596	0.214
Logistic regression	0.582	0.193
Gradient boosting	0.670	0.290
MM-Net (proposed)	0.782	0.419
(b) MM-Net AUC by event type		
Event type	Positive epochs	AUC↑
Hypopnea	11,605	0.748
Obstructive	1,661	0.898
Central	831	0.935

I. CLINICAL VALIDATION AGAINST AHI AND SEVERITY

Aggregating the per-epoch respiratory probability into a per-patient burden and correlating it with the clinical apnea-hypopnea index (AHI) from the metadata gives a significant positive association (Spearman $\rho = 0.491$, $p < 0.001$, $n = 99$, the three recordings with an incomplete cardiorespiratory montage scored with those channels zero-filled; Fig. 8), so the model's output tracks a clinician's summary measure, not only an epoch-level score. Staging, in turn, degrades as sleep-disordered breathing worsens: accuracy falls from 0.769 in patients with a normal AHI to 0.719 in the severe group (Table 8). The pathology that motivates the second output also makes the first harder, which is the clinical reality of this cohort.

The association is real but modest, and it is worth saying why rather than only that it is so. Three mechanisms plausibly cap it, and each is a property of the formulation rather than of the model. First, mean per-epoch probability is a poor estimator of an hourly event rate: a 30-second epoch holding one hypopnea and one holding three both contribute a single positive, so the aggregate compresses exactly the variation the index is built from. Second, the clinical index is normalised by total sleep time, whereas our burden averages over every

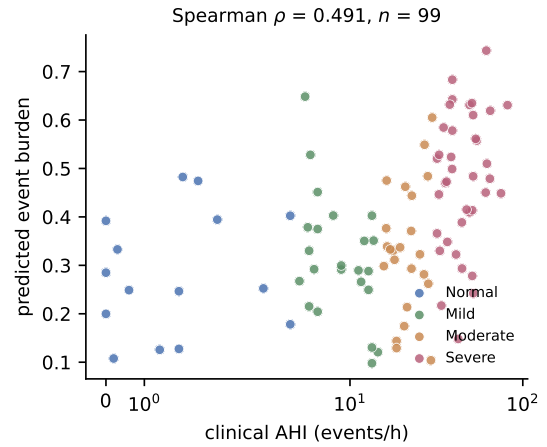


FIGURE 8. Predicted per-patient event burden (mean respiratory probability) against the clinical apnea-hypopnea index, coloured by AASM severity band. The positive association (Spearman $\rho = 0.491$, $n = 99$) shows the model's output tracks a clinician's summary measure.

TABLE 8. Sleep-staging accuracy by sleep-disordered-breathing severity (AASM AHI bands), ten-fold patient-independent; mean $\pm$ SD across patients ($N = 99$). Staging degrades monotonically as severity rises.

AHI band	AHI (events/h)	Patients	Staging acc ↑
Normal	< 5	15	0.769 $\pm$ 0.049
Mild	5–15	23	0.747 $\pm$ 0.069
Moderate	15–30	23	0.722 $\pm$ 0.088
Severe	≥ 30	38	0.719 $\pm$ 0.100

scored epoch including wake, and the two denominators diverge most in the patients with the most fragmented sleep, which in this cohort is the severe group. Third, the AHI itself carries substantial night-to-night variability, so a single-night reference sets a ceiling on any correlation with it. We therefore read $\rho = 0.491$ as evidence that the read-out tracks severity at the group level, and not as a claim that it estimates an index.

J. PERMUTATION IMPORTANCE: A SECOND, INDEPENDENT ATTRIBUTION

The ablation grid establishes what the model *needs*: remove a modality, retrain, and observe which head suffers. A complementary question is what a *trained* model actually uses. We therefore permute each modality's features across epochs at inference time, so the channel is still present but no longer aligned to the epoch it describes, and measure the drop, averaged over three shuffles and the ten fold models.

The two methods can disagree, because a model may need a channel during training yet weight it lightly afterwards. On staging they do not: permuting the neural features costs 0.284 staging accuracy against 0.030 for the ocular and 0.006 for the muscle channel, the same ordering the ablation gives, and the rank correlation between the two attributions is $\rho = 0.833$ ($p = 0.010$).

On the respiratory head the agreement is narrower, and we

state it precisely rather than pooling the two into a single coefficient. Both methods place respiratory effort first by a wide margin: 0.038 AUC under retraining, 0.071 under permutation, in each case well clear of the runner-up. The remaining four cardiorespiratory channels, however, have retraining effects that are not distinguishable from zero, so their rank order carries no signal and the correlation over the five is correspondingly uninformative ($\rho = 0.381$, $p = 0.35$). The claim the comparison supports is therefore the one that matters physiologically: two attributions sharing no machinery (one applied to retrained models, one to fixed ones), independently identify effort as the dominant respiratory cue. Unlike an embedding projection, that was a prediction that could have failed.

K. DATA EFFICIENCY: WHICH CEILING ARE WE AGAINST?

Staging on this corpus saturates near 0.72–0.75 for every model family tried here and elsewhere. Whether that reflects the architecture or the cohort is testable: retrain the identical model on a shrinking fraction of the *training patients*, always evaluating on the full held-out fold (Fig. 9). Patients are subsampled, not epochs, because epochs within a patient are heavily correlated.

The two heads answer differently. Staging accuracy rises 0.701 $\rightarrow$ 0.721 $\rightarrow$ 0.726 $\rightarrow$ 0.735 as training patients go from 20 to 79, with marginal gains of +0.020, +0.005 and +0.009, all under half a fold standard deviation and flattening. Respiratory AUC over the same range rises 0.732 $\rightarrow$ 0.750 $\rightarrow$ 0.752 $\rightarrow$ 0.785, and its *largest* increment is the final one, +0.033, close to a full fold standard deviation, and steeper than any earlier step. This curve is a single seed on a fixed held-out set, so its endpoint differs slightly from the three-seed headline figures (0.739 and 0.782); the shape, not the endpoint, is what it is here to show. **Staging is at or near this cohort's ceiling; respiratory detection is not, and is still accelerating.** We draw the narrower conclusion the data support: more patients of this kind would improve breathing detection and would not appreciably improve staging, so the staging ceiling is a property of the labels and the injured brain, not of the training-set size. The final model also exceeds the feature-only configuration at the full training-set size, so its advantage is not an artefact of the full sample.

L. HEALTHY-TRAINED MODELS APPLIED TO THE INJURED BRAIN

Table 4 retrains published architectures on iSLEEPS. A different and harder question is what happens when a model built for healthy sleep is applied to this cohort as published. The three architectures are ours, not published models: a four-channel CNN, the same CNN with a BiLSTM over epochs, and StagingSeqNet, a deeper convolutional-recurrent stager we built for this comparison. Naming them matters only in that none is a weak straw man; each reaches 0.81 to 0.87 on the healthy corpus it was trained on. We trained them on the Sleep-EDF Cassette subset and applied each to all 99 iSLEEPS patients zero-shot, with no adaptation of any kind.

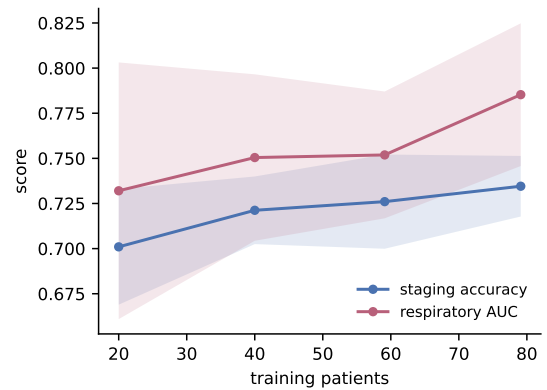


FIGURE 9. Learning curve over training-patient fraction, ten-fold patient-independent, error bars one standard deviation across folds. Staging (blue) flattens by 57 patients; respiratory detection is still climbing at 76. Bands are $\pm$ one standard deviation across folds.

This is the middle block of Table 4, set against the same architectures retrained on this cohort.

The result is not a degradation but a collapse. All three stage healthy sleep competently, 0.806 to 0.868 accuracy and κ 0.73 to 0.83; on stroke the same checkpoints reach 0.307 to 0.325 with κ 0.09 to 0.11, which is close to chance for a five-class problem with this prior. The failure mode is more informative than the headline: each model labels 66–78% of stroke epochs Wake against a true 26.7%, and N2 recall falls to 0.032 on the stage that is 42% of the data. These models do not lose accuracy gracefully; they stop recognising sleep architecture at all.

The control is what makes this interpretable. A healthy-trained stager scoring 0.31 on iSLEEPS is evidence of a cohort effect only if the same checkpoint, through the same preprocessing, scores properly on healthy sleep; otherwise a broken input pipeline and a genuine domain gap produce identical numbers. Run through our pipeline on held-out Sleep-EDF recordings, the strongest checkpoint holds at 0.8682 (0.8667 ± 0.0415 per recording), so the pipeline is correct and the collapse belongs to the cohort.

Two bounds, reported together. Sleep-EDF is recorded on Fpz-Cz and Pz-Oz where iSLEEPS uses C4:M1 and O2:M1, so part of this gap is montage shift, not pathology, and quoting the zero-shot number alone would overstate the case. Deep models trained on iSLEEPS with the matched montage reach 0.61–0.69 (upper block), which puts the cost of pathology alone at roughly 0.19 against healthy-sleep performance; healthy-trained and zero-shot, with the montage shift included, the cost is 0.56. The honest claim is bounded by those two figures rather than by either alone.

The same argument run in reverse. Every figure above concerns other architectures, which leaves the obvious question unanswered: whether 0.739 is low because the cohort is hard or because the proposed model is weak. So we cross-validated it on healthy sleep, ten-fold and subject-independent over the 21 Sleep-EDF subjects, using the neural branch alone since that corpus has no cardiorespiratory

channels. It reaches 0.856 ± 0.027 accuracy with κ 0.800 (final row of Table 4). That places it among the healthy-sleep stagers in the middle block, not below them, against 0.741 for the identical architecture on stroke. The 0.115 difference is what the cohort costs a model that is demonstrably capable of more, which is a stronger statement than any comparison against other people's architectures can make.

The cohort does not cost every architecture equally. Because the three reimplemented models carry a healthy-sleep score produced by the same code under the same protocol, the table supports a comparison it could not before: not merely where each model lands, but how much it loses in the crossing. TinySleepNet falls 0.148 from healthy sleep to stroke ($0.824 \rightarrow 0.676$), SleepTransformer 0.140 ($0.823 \rightarrow 0.683$) and U-Time 0.145 ($0.791 \rightarrow 0.646$), against 0.115 for the proposed model ($0.856 \rightarrow 0.741$). It is therefore not simply higher at both ends; it degrades least. We report this as an observation and not a claim. Four paired differences will not support a significance test, and the comparison is between systems and not a controlled ablation, since the proposed rows read engineered features and a frozen embedding over four derivations while the three reimplemented baselines read a single raw channel. The direction is nonetheless the one the rest of the paper predicts: a representation that encodes sleep physiology explicitly survives a disordered cortex better than one a network must discover for itself from fewer than a hundred patients.

Folds are split by subject, not by recording, because Sleep-EDF Cassette records each subject on two nights; splitting on filenames would place the same person on both sides of a fold.

M. WHAT EACH STREAM SHOULD BE REPRESENTED AS

The architecture makes two representation choices (engineered features plus a frozen self-supervised embedding on the neural side, a learned convolutional encoder over the raw signal on the cardiorespiratory side), and both were selected by experiment, not assumed. Each family below holds every other hyperparameter fixed and varies one thing, over the same three seeds and ten patient-independent folds, with Holm correction within the family.

The cardiorespiratory features are the bottleneck, and part of the gap needs no new information. Each block of Table 9 holds the other stream at its baseline, so no single row there is the full model. The upper block replaces the 14 engineered cardiorespiratory features with progressively richer representations of the same underlying signal. Every step is significant, and the ordering is instructive. A random nonlinear expansion of the same 14 numbers recovers 0.018 AUC: that gain needs no new measurement at all, only more capacity to read what was already computed, which says the summary features leave nonlinear structure unextracted. A random linear projection of the raw waveform adds a further 0.016, so part of what the features discard is recoverable without learning anything. The learned convolutional encoder reaches 0.782, 0.084 above the engineered features.

What this establishes is that the representation of the cardiorespiratory stream, not the detector on top of it, is what limited the respiratory read-out. We therefore claim the change of representation, not the design of the encoder that implements it.

On the neural side a pretrained encoder does not replace expert features, but does add to them. The lower block of Table 9 varies the neural representation with the cardiorespiratory stream held at the engineered features. Used alone, the frozen encoder is significantly *worse* than the 188 features (-0.028 , Holm $p < 0.001$): pretraining on healthy scalp EEG does not by itself substitute for physiology chosen for this task. Concatenated with the features it adds $+0.008$ accuracy (Holm $p = 0.001$), and a second pretrained encoder on top of the first adds nothing further (-0.004 , $p = 0.24$).

The reading we take is narrow, because it is what the numbers support: expert physiology and self-supervised pre-training encode partly different information, so their union beats either alone; but the complementarity saturates after one pretrained encoder, so this is not a general argument for stacking foundation models.

N. CALIBRATION AND WHEN THE MODEL DECLINES TO COMMIT

A clinical reader needs to know not only how often the model is right but whether it knows when it is likely to be wrong. Two measurements answer that, both computed on held-out epochs with the raw posteriors before HMM smoothing.

The model is *overconfident*: expected calibration error is 0.076, and the gap runs one way in every confidence bin: epochs it scores above 0.9 are correct 88% of the time. Stated confidences should therefore be read as ordered, not as probabilities.

Ranking, however, is what abstention needs. Applying split-conformal prediction with the fold's validation patients as the calibration set gives empirical coverage that tracks the target closely (0.894 at a 0.90 target, 0.943 at 0.95). At $\alpha = 0.10$ the model returns a single stage for 45.8% of epochs and is correct on 86.9% of those, against 59.1% on the epochs where it returns a larger set. The abstention is therefore selecting the genuinely hard epochs and not merely hedging, and it concentrates where a scorer would expect: N1 receives a singleton set only 19% of the time, the lowest of any stage and the one human scorers agree on least.

O. EXTERNAL VALIDATION ON TWO INDEPENDENT CORPORA

Ten-fold cross-validation is patient-independent but remains within one corpus, one centre and one acquisition protocol. We therefore trained a single model on all 99 iSLEEPS patients and evaluated it, unchanged, on two public corpora with no dataset-specific tuning: no fine-tuning, no threshold search, no re-fitted model or decision threshold (per-recording feature standardisation is applied, as in training), and hidden-Markov transitions estimated from iSLEEPS alone (Table 10).

TABLE 9. What each stream should be represented as. One factor varies per block with everything else held fixed; ten-fold patient-independent, three seeds, thirty fold-values per cell, mean $\pm$ SD. * and Holm p are against the engineered-feature baseline of each block, Wilcoxon paired on fold-values. "adopted" marks the choice carried into the final model.

Representation	Variant	Resp. AUC $\uparrow$	Staging acc $\uparrow$	Δ	Holm p
Cardiorespiratory	14 engineered features	0.698 ± 0.051	0.736 ± 0.014	—	—
	random nonlinear expansion of the 14	0.715 ± 0.047	0.737 ± 0.018	+0.018	< 0.001
	random linear projection of raw signal	0.731 ± 0.044	0.735 ± 0.021	+0.034	< 0.001
	learned CNN, 0.155 M (adopted)	0.782 ± 0.040	0.740 ± 0.019	+0.084	< 0.001
Neural	188 engineered features	0.685 ± 0.041	0.741 ± 0.016	—	—
	frozen pretrained encoder alone	0.681 ± 0.033	0.712 ± 0.033	−0.028	< 0.001
	features + pretrained (adopted)	0.690 ± 0.037	0.749 ± 0.018	+0.008	0.001
	features + two pretrained encoders	0.684 ± 0.043	0.745 ± 0.018	+0.004	0.23

The two corpora test different things. **Sleep-EDF** [40] is healthy sleep recorded on a frontal montage (Fpz-Cz, Pz-Oz) with no cardiorespiratory channels, so it tests the staging head across both a cohort shift and a montage shift. **ISRUC-Sleep** [49] is a sleep-disorders cohort recorded on the same central and occipital derivations we use (C4, C3, O2, O1 against the contralateral mastoid), with SpO₂, airflow, effort and ECG, and with obstructive, central and mixed apneas and hypopneas scored, so it tests the staging head *without* a montage confound and provides the only external test of the respiratory head. Because Sleep-EDF carries no cardiorespiratory channels, that branch is fed zeros there. Zeroing an input the network trained on is a train/test mismatch, not an ablation, so we also trained a model with the branch zeroed throughout: it reaches 0.774 accuracy against the 0.794 reported. We read the pair as bracketing the true transfer figure, not as confirming the higher one. The mismatched run could be gaining from batch-normalisation statistics that never saw zeros, in which case 0.774 is the more defensible number, and a reader who prefers it loses nothing in our argument: both sit above the in-domain 0.739. Both corpora also differ from iSLEEPS in montage, so both transfer figures are lower bounds rather than clean estimates. ISRUC records each subject twice; respiratory-event prevalence is 10.1% on the first night and 2.4% on the second, so we report the two separately instead of pooling a difference that has nothing to do with the model.

The respiratory read-out transfers. On ISRUC night one the model reaches 0.780 AUC against 0.782 in-domain, a gap of 0.002, and it does so handicapped, because ISRUC provides no pulse channel and no separate effort trace, leaving two of the seven cardiorespiratory inputs zero-filled. A detector that had latched onto one centre's equipment would not survive that; this one does. Night two falls to 0.723, tracking its much lower event prevalence (2.4% against 10.1%) and not anything about the model.

Staging moves in opposite directions on the two corpora, for reasons the data identify. On Sleep-EDF the model *gains* 0.083 of κ over its in-domain score. A representation that had memorised corpus-specific artefacts would degrade out of corpus, so the straightforward reading is that the features are physiological and that the ceiling reported

throughout this paper is a property of *stroke sleep* rather than of the model. The learning curve reaches the same conclusion (Section VI-K) by a different route. On ISRUC staging falls by 0.090 accuracy, and the class mix explains much of it: ISRUC carries 15.6% N1 against iSLEEPS' 10.2% and Sleep-EDF's 6.6%, and N1 is the weakest stage in every cohort we have measured (recall 0.168 on ISRUC, 0.294 on Sleep-EDF, and the lowest per-class F1 in-domain).

A montage explanation the data now rule out. An earlier version of this work observed N3 recall of 0.554 on Sleep-EDF with 44% of N3 epochs read as N2, and attributed it to slow-wave activity being frontally dominant while the model reads central and occipital derivations. That account made a prediction: N3 should recover on ISRUC, which uses our derivations exactly. Under the final model the prediction fails, and it fails in the direction that rules the explanation out. N3 recall is 0.841 on Sleep-EDF, the mismatched frontal montage, with 12.7% of N3 read as N2, and 0.630 on ISRUC, the matching one, with 34.8%. Electrode position is therefore not what governs N3 transfer here. The only component that changed between the two models is the neural representation (Section VI-M), which points instead at how the cortex is encoded, though with one external pass per corpus and eight ISRUC subjects we can identify the effect without attributing it confidently. We keep the failed prediction on the record because it was falsifiable, it was tested on a corpus chosen for an unrelated reason, and it was wrong.

Two limits keep this in proportion. Accuracy across cohorts with different class balance is not directly comparable, which is why κ is reported alongside. And ISRUC contributes only eight subjects. Per-recording accuracy spans 0.458–0.748, so the ISRUC figures in Table 10 are quoted to three decimals for consistency with the rest of the paper and should be read to two; per-recording respiratory AUC spans 0.515–0.848, the latter tracking event prevalence closely, so the low-prevalence nights give near-chance estimates.

P. MODEL COMPLEXITY AND CAPABILITY

Table 11 sets the proposed model beside representative alternatives on size and capability. At 2.76 M trainable parameters the model is an order of magnitude smaller than DeepSleep-Net (~21 M) and the corpus's own CNN-ResNet18 (~11 M), and comparable to SleepTransformer, so size is not where it

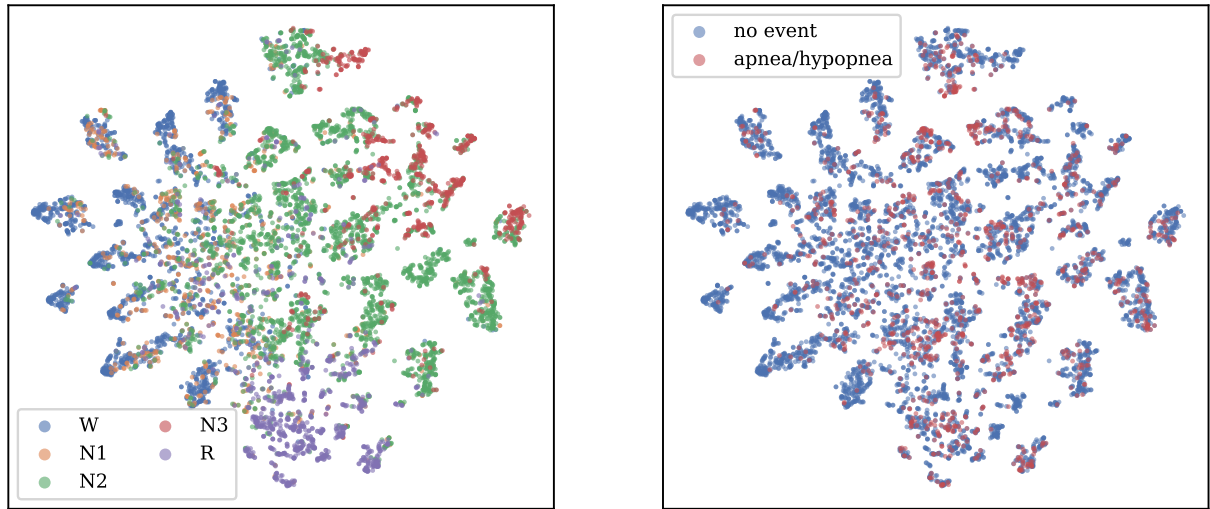


FIGURE 10. Two-dimensional t-SNE of the model's per-epoch embeddings (pooled test epochs). Left: by sleep stage; Wake, N2, N3 and REM form distinct territories while N1 is diffuse. Right: by respiratory-event label; events spread across the manifold, consistent with the modest detection AUC.

TABLE 10. External validation on two public corpora, zero-shot: one model trained on all 99 iSLEEPS patients with no dataset-specific tuning. In-domain figures are ten-fold patient-independent means. Sleep-EDF carries no cardiorespiratory channels, so the respiratory head is not scored there. The two ISRUC nights differ sharply in event prevalence and are reported separately.

Corpus	Cohort	Rec.	Acc $\uparrow$	κ $\uparrow$	Resp. AUC $\uparrow$
iSLEEPS (in-domain)	stroke	99	0.739	0.634	0.782
Sleep-EDF	healthy	40	0.794	0.718	—
ISRUC, night 1	sleep-disord.	8	0.649	0.536	0.780
ISRUC, night 2	sleep-disord.	8	0.575	0.426	0.723

TABLE 11. Model size and clinical capability. The proposed model is mid-sized and is the only one that produces both a stage and a respiratory-event label.

Model	Input	Params $\downarrow$	Outputs
CNN-ResNet18 [13]	1 EEG	~ 11 M	stage
DeepSleepNet [2]	1 EEG	~ 21 M	stage
Feature-sequence BiLSTM	7 ch feat	1.2 M	stage
Raw multimodal CNN	14 ch	0.95 M	stage + apnea
MM-Net (proposed)	7 ch feat + raw	2.76 M	stage + apnea

wins. It delivers both a competitive stage and a respiratory read-out from one pass over the recording, and none of the staging-only models, at any size, can supply the second output.

VII. DISCUSSION

The result is a single model that reads the whole sleeping body and speaks to two clinical questions at once, in the population where both matter. The learned per-epoch embedding (Fig. 10) separates cleanly by sleep stage while the respiratory structure stays diffuse, mirroring the strong staging and the modest respiratory AUC.

Staging accuracy on this cohort saturates below 0.75 for

every model family we evaluated, and the proposed model reaches 0.739 while adding a respiratory capability the staging models lack. The respiratory read-out does not depend on the labeling channel: it survives deletion of the airflow channel and rests on respiratory effort, desaturation, and arousal, the same signs a physician reads at the bedside.

The 0.739 staging figure should be judged on this cohort, where healthy-sleep accuracy does not transfer. Deep architectures developed on healthy sleepers collapse here: DeepSleepNet reaches 0.615, CNN-ResNet18 0.617, and a network pretrained on the healthy Sleep-EDF corpus and applied to this cohort 0.623, each roughly fifteen to twenty points below its healthy accuracy of about 0.85. The injured brain distorts the very micro-events, spindles, slow waves, and atonia, that those models learned to read. What holds up is physiologically grounded modeling on engineered features, at 0.72 to 0.75, which matches the published baseline on this corpus. The lesson is that healthy-sleep deep learning fails on the injured brain, and this cohort requires cohort-specific, physiologically grounded, multimodal modeling.

The study also draws a clean physiological line. The cortical channels carry sleep staging, and the cardiorespiratory channels carry breathing, and the model places each where it belongs rather than blurring them. That the neural and multimodal variants stage alike is the expected outcome: the second modality adds a new capability while leaving staging unchanged. We view this separation as a strength for clinical use, because a reader can trust that the respiratory output reflects breathing and the staging output reflects the brain.

A. LIMITATIONS

The system is offline, and mildly transductive. Two properties follow from the design and neither is visible in the numbers. Staging uses whole-night Viterbi decoding, so no epoch can be scored until the recording ends, which rules out

the real-time use a bedside monitor would need. And features are standardised per patient, which at test time uses that night's own statistics; the model never sees another patient's labels, so this is not label leakage, but it is a transductive step and a strictly causal deployment would need running or population statistics instead. We have not measured what that substitution costs.

A prediction of ours that failed. The montage account of N3 transfer described in Section VI-O does not survive retesting under the final model, and we report it as falsified instead of removing it. Two cautions follow for every other per-class figure in that section: each rests on a single external pass with no fold structure, so none carries an interval, and ISRUC contributes eight subjects, at which size a per-class rate is an estimate with a width we cannot quote. Per-class transfer should be read as direction, not magnitude.

The evidence comes from a single center and the 96 patients with complete cardiorespiratory channels. Section VI-O shows both heads transfer to independent corpora without tuning, but neither external corpus is a stroke cohort, so every stroke-specific claim rests on iSLEEPS alone; multi-center validation on a second stroke cohort is required before clinical use, and no such public corpus currently exists. The external respiratory result rests on eight subjects, two of whose seven cardiorespiratory inputs were unavailable, so it establishes that the read-out transfers, not how well. Respiratory detection, at 0.782 AUC, is a useful signal but not yet a screening-grade one, and it is scored at the epoch level, not per event. At event level it detects 55% of events at 2.6 false alarms per hour and under-counts the event rate by more than half, so it cannot substitute for a scored index. The staging head is also overconfident (expected calibration error 0.076), so its posteriors should be used for ranking and abstention, not read as probabilities. Staging sits at the cohort ceiling, so the model does not raise staging accuracy above the published baseline for this corpus. Finally, the model is feature-based: it inherits the assumptions of the engineered representations and is only partly an end-to-end raw-signal learner, since the neural stream reads engineered descriptors and a frozen embedding while the cardiorespiratory stream is trained on the raw signal, a deliberate trade for data efficiency on a small cohort.

B. FUTURE WORK

Two directions follow naturally. Event-level metrics, such as sensitivity per event and false alarms per hour, would sit beside the epoch-level AUC and speak more directly to a sleep technician, and finer modeling of the breathing waveforms is likely to raise the respiratory number further. Linking the multimodal representation to stroke severity and recovery is the larger opportunity, and one the cardiorespiratory signal is well placed to serve.

VIII. CONCLUSION

We built a model that reads the full polysomnogram of stroke patients and, from one pass, stages their sleep and detects

their respiratory events. Deep networks built for healthy sleep collapse to 0.61–0.69 accuracy on this cohort, so its 0.739 staging accuracy is at the cohort ceiling for every model family. The contribution is the second output, a respiratory read-out at 0.782 AUC that survives removal of the airflow channel, showing it reads the underlying physiology. Reading the whole recording gives the model a second clinical voice in the population that needs it most. Code and features are released.

DATA AND CODE AVAILABILITY

The iSLEEPS corpus is publicly available (Zenodo, DOI 10.5281/zenodo.14873844; and the india-data.org portal) as described by Maiti *et al.* [13]. Our code, engineered features, and reproduction notebook are released at <https://github.com/wshuv-o/isleeps-sleep-staging>.

ETHICS

The iSLEEPS data collection was approved by the NIMHANS Institutional Ethics Committee (No. NIMHANS/34th IEC (BS&NS DIV.)/2022, dated 05.02.2022) [13]. This work is a secondary analysis of the de-identified public release and required no additional enrollment.

FUNDING

The authors received no specific funding for this work.

ACKNOWLEDGMENT

The authors thank the iSLEEPS investigators for releasing the corpus.

REFERENCES

- [1] American Academy of Sleep Medicine, *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications, Version 3*. American Academy of Sleep Medicine, 2023.
- [2] A. Supratak, H. Dong, C. Wu, and Y. Guo, "DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel EEG," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 11, pp. 1998–2008, 2017. [Online]. Available: <https://doi.org/10.1109/TNSRE.2017.2721116>
- [3] H. Phan, F. Andreotti, N. Cooray, O. Y. Chén, and M. De Vos, "SeqSleepNet: End-to-end hierarchical recurrent neural network for sequence-to-sequence automatic sleep staging," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 27, no. 3, pp. 400–410, 2019. [Online]. Available: <https://doi.org/10.1109/TNSRE.2019.2896659>
- [4] A. Supratak and Y. Guo, "TinySleepNet: An efficient deep learning model for sleep stage scoring based on raw single-channel EEG," in *42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 2020, pp. 641–644. [Online]. Available: <https://doi.org/10.1109/EMBC44109.2020.9176741>
- [5] E. Eldele, Z. Chen, C. Liu, M. Wu, C.-K. Kwok, X. Li, and C. Guan, "An attention-based deep learning approach for sleep stage classification with single-channel EEG," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 809–818, 2021. [Online]. Available: <https://doi.org/10.1109/TNSRE.2021.3076234>
- [6] H. Phan, O. Y. Chén, M. C. Tran, P. Koch, A. Mertins, and M. De Vos, "XSleepNet: Multi-view sequential model for automatic sleep staging," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 9, pp. 5903–5915, 2022. [Online]. Available: <https://doi.org/10.1109/TPAMI.2021.3070057>

TABLE 12. Recent single-channel sleep-staging methods: the accuracy each reports on healthy or general cohorts, set against its accuracy on the subacute-stroke cohort (iSLEEPS) for the five architectures we evaluated on it. A dash marks a value we did not evaluate.

Method	Year	Approach	Reported acc (dataset)	On iSLEEPS
DeepSleepNet [2]	2017	Two representation-learning CNNs followed by a bidirectional LSTM sequence model, single-channel EEG.	0.82 (Sleep-EDF)	0.615
U-Time [7]	2019	Fully convolutional encoder-decoder segmenting a whole night at once, no recurrence over epochs.	≈0.79 (Sleep-EDF)	0.646
TinySleepNet [4]	2020	Compact convolutional stack with a single unidirectional LSTM, single-channel EEG.	0.854 (Sleep-EDF)	0.676
AttnSleep [5]	2021	Multi-resolution CNN with multi-head self-attention over epochs, single-channel EEG.	0.844 (Sleep-EDF)	0.686
SleepTransformer [17]	2022	Transformer encoder with epoch-level interpretability and uncertainty estimates.	0.848 (Sleep-EDF)	0.683
1D-ResNet-SE-LSTM	2023	Residual network with squeeze-and-excitation blocks and an LSTM on raw EEG.	0.864 (Sleep-EDF)	–
Micro SleepNet	2023	Lightweight convolutional model for real-time mobile staging.	0.833 (SHHS)	–
AT-BiLSTM	2024	Attention-augmented bidirectional LSTM, single-channel EEG.	0.838 (Sleep-EDF)	–
MM-Net (proposed)	–	Two feature streams (neural and cardiorespiratory) fused into a BiLSTM that jointly stages sleep and detects respiratory events from the full polysomnogram.	–	0.739

- [7] M. Perslev, M. H. Jensen, S. Darkner, P. J. Jennum, and C. Igel, “U-Time: A fully convolutional network for time series segmentation applied to sleep staging,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 4417–4428.
- [8] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, “Large brain model for learning generic representations with tremendous EEG data in BCI,” in *International Conference on Learning Representations (ICLR)*, 2024, arXiv:2405.18765.
- [9] J. Wang, S. Zhao, Z. Luo, Y. Zhou, H. Jiang, S. Li, T. Li, and G. Pan, “CBraMod: A criss-cross brain foundation model for EEG decoding,” in *International Conference on Learning Representations (ICLR)*, 2025, arXiv:2412.07236.
- [10] G. Wang, W. Liu, Y. He, C. Xu, L. Ma, and H. Li, “EEGPT: Pretrained transformer for universal and reliable representation of EEG signals,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024. [Online]. Available: <https://doi.org/10.52202/079017-1239>
- [11] C. Yang, M. B. Westover, and J. Sun, “BIOT: Biosignal transformer for cross-data learning in the wild,” in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 78 240–78 260. [Online]. Available: <https://openreview.net/forum?id=c2LZyTyddi>
- [12] H. Phan and K. Mikkelsen, “Automatic sleep staging of EEG signals: Recent development, challenges, and future directions,” *Physiological Measurement*, vol. 43, no. 4, 2022. [Online]. Available: <https://doi.org/10.1088/1361-6579/ac6049>
- [13] S. Maiti, S. K. Sharma, S. Mythirayee, S. Rajendran, and R. S. Bapi, “Polysomnography dataset for sleep analysis in ischemic stroke patients,” *Scientific Data*, vol. 13, no. 1, p. 421, 2026. [Online]. Available: <https://doi.org/10.1038/s41597-026-06747-w>
- [14] C. L. Bassetti and M. S. Aldrich, “Sleep electroencephalogram changes in acute hemispheric stroke,” *Sleep Medicine*, vol. 2, no. 3, pp. 185–194, 2001. [Online]. Available: [https://doi.org/10.1016/s1389-9457\(00\)00071-x](https://doi.org/10.1016/s1389-9457(00)00071-x)
- [15] C. Denis, I. Jaussent, L. Guiraud, C. Mestejanot, C. Arquizan, I. Mourand, S. Chenini, B. Abril, A. Wacongne, R. Tamisier, S. Baillieux, J.-L. Pepin, L. Barateau, and Y. Dauvilliers, “Functional recovery after ischemic stroke: Impact of different sleep health parameters,” *Journal of Sleep Research*, vol. 33, no. 1, 2024. [Online]. Available: <https://doi.org/10.1111/jsr.13964>
- [16] S. Bose, S. Maiti, S. K. Sharma, S. Mythirayee, T. Chakraborti, S. Rajendran, and R. S. Bapi, “AI generalisation gap in comorbid sleep disorder staging,” in *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2026. [Online]. Available: <https://doi.org/10.1109/ISBI61048.2026.11515484>
- [17] H. Phan, K. Mikkelsen, O. Y. Chén, P. Koch, A. Mertins, and M. De Vos, “SleepTransformer: Automatic sleep staging with interpretability and uncertainty quantification,” *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 8, pp. 2456–2467, 2022. [Online]. Available: <https://doi.org/10.1109/TBME.2022.3147187>
- [18] H. Phan, K. P. Lorenzen, E. Heremans, O. Y. Chén, M. C. Tran, P. Koch, A. Mertins, M. Baumert *et al.*, “L-SeqSleepNet: Whole-cycle long sequence modelling for automatic sleep staging,” *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 10, pp. 4748–4757, 2023. [Online]. Available: <https://doi.org/10.1109/JBHI.2023.3303197>
- [19] N. Jirakittayakorn, Y. Wongsawat, and S. Mitirattanakul, “ZleepAnlyst-Net: A novel deep learning model for automatic sleep stage scoring based on single-channel raw EEG data using separating training,” *Scientific Reports*, vol. 14, no. 1, p. 9859, 2024. [Online]. Available: <https://doi.org/10.1038/s41598-024-60796-y>
- [20] Y. Guo, M. Nowakowski, and W. Dai, “FlexSleepTransformer: A transformer-based sleep staging model with flexible input channel configurations,” *Scientific Reports*, vol. 14, no. 1, p. 26312, 2024. [Online]. Available: <https://doi.org/10.1038/s41598-024-76197-0>
- [21] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.
- [22] X. Zhou, Y. Han, Z. Chen, C. Liu, Y. Ding, Z. Jia, and Y. Liu, “BiT-MamSleep: Bidirectional temporal mamba for EEG sleep staging,” *arXiv preprint arXiv:2411.01589*, 2024. [Online]. Available: <https://doi.org/10.48550/ARXIV.2411.01589>
- [23] M. Perslev, S. Darkner, L. Kempfner, M. Nikolic, P. J. Jennum, and C. Igel, “U-Sleep: Resilient high-frequency sleep staging,” *npj Digital Medicine*, vol. 4, no. 1, p. 72, 2021.
- [24] R. Thapa, M. R. Kjaer, B. He, I. Covert, H. Moore IV, U. Hanif *et al.*, “A multimodal sleep foundation model for disease prediction,” *Nature Medicine*, 2026. [Online]. Available: <https://doi.org/10.1038/s41591-025-04133-4>
- [25] Y. Zhou, J. Wu, Z. Ren, Z. Yao, W. Lu, K. Peng, Q. Zheng, C. Song *et al.*, “CSBrain: A cross-scale spatiotemporal brain foundation model for EEG decoding,” *arXiv preprint arXiv:2506.23075*, 2025. [Online]. Available: <https://doi.org/10.52202/085713-2915>
- [26] K. Kontras, T. Osselaer, S. G. Mouslech, A.-I. Karaiskou, G. Gagliardi, T. Strypsteen *et al.*, “NeuroAtlas: Benchmarking foundation models for clinical EEG and brain-computer interfaces,” *arXiv preprint arXiv:2605.14698*, 2026.
- [27] A. Dei Rossi, M. Metaldi, M. Bechny, I. Filchenko, J. van der Meer, M. H. Schmidt, C. L. A. Bassetti, A. Tzovara *et al.*, “SLEEPYLAND: Trust begins with fair evaluation of automatic sleep staging models,” *npj Digital Medicine*, vol. 9, no. 1, p. 55, 2026. [Online]. Available: <https://doi.org/10.1038/s41746-025-02237-2>

- [28] E. Estevan, M. Sierra-Torralba, E. López-Larraz, and L. Montesano, "A systematic evaluation of self-supervised learning for label-efficient sleep staging with wearable EEG," *arXiv preprint arXiv:2510.07960*, 2025. [Online]. Available: <https://doi.org/10.48550/ARXIV.2510.07960>
- [29] A. Guillot and V. Thorey, "RobustSleepNet: Transfer learning for automated sleep staging at scale," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 1441–1451, 2021. [Online]. Available: <https://doi.org/10.1109/TNSRE.2021.3098968>
- [30] K. Yang and H. Wang, "Deep learning for EEG-based sleep stage classification: A review," *Medical & Biological Engineering & Computing*, vol. 64, pp. 2017–2043, 2026. [Online]. Available: <https://doi.org/10.1007/s11517-026-03583-3>
- [31] U. Hanif, A. Aloulou, F. Crosbie, P. Bouchequet, M. Chennaoui, T. Andriillon, and D. Leger, "Deciphering insomnia: Benchmarking automated sleep staging algorithms for complex sleep disorders," *Journal of Sleep Research*, vol. 34, no. 6, p. e70048, 2025. [Online]. Available: <https://doi.org/10.1111/jsr.70048>
- [32] T. Jing, L. Xu, S. Sun, W. Yu, Y. Zheng, G. Xu, X. Shen, T. Tang, and C. Chu, "CPAP modulation of sleep microstructure as predictors of neurological recovery in ischemic stroke patients with moderate-to-severe OSA," *Frontiers in Neurology*, vol. 16, p. 1605106, 2025. [Online]. Available: <https://doi.org/10.3389/fneur.2025.1605106>
- [33] N. Tellenbach, M. H. Schmidt, F. Alexiev, E. Blondiaux, F. Cavalloni, C. L. Bassetti, L. Heydrich, and P. Bargiotas, "REM sleep and muscle atonia in brainstem stroke: A quantitative polysomnographic and lesion analysis study," *Journal of Sleep Research*, vol. 32, no. 2, 2022.
- [34] J. M. Gottselig, C. L. Bassetti, and P. Achermann, "Power and coherence of sleep spindle frequency activity following hemispheric stroke," *Brain*, vol. 125, no. 2, pp. 373–383, 2002. [Online]. Available: <https://doi.org/10.1093/brain/awf021>
- [35] L. M. J. Fernández and A. Lüthi, "Sleep spindles: Mechanisms and functions," *Physiological Reviews*, vol. 99, no. 2, pp. 1349–1385, 2019. [Online]. Available: <https://doi.org/10.1152/physrev.00042.2018>
- [36] B. K. Simpson, R. Rangwani, A. Abbasi, J. M. Chung, C. M. Reed, and T. Gulati, "Disturbed laterality of non-rapid eye movement sleep oscillations in post-stroke human sleep: A pilot study," *Frontiers in Neurology*, vol. 14, p. 1243575, 2023. [Online]. Available: <https://doi.org/10.3389/fneur.2023.1243575>
- [37] L. Facchin, C. Schöne, A. Mensen *et al.*, "Slow waves promote sleep-dependent plasticity and functional recovery after stroke," *Journal of Neuroscience*, vol. 40, no. 45, pp. 8637–8651, 2020. [Online]. Available: <https://doi.org/10.1523/jneurosci.0373-20.2020>
- [38] Y. Zheng, T. Jing, L. Xu, S. Sun, W. Yu, R. Liu, G. Xu, and C. Chu, "Analysis of the correlation between sleep spindles and cognitive impairment in patients with ischemic stroke," *Scientific Reports*, vol. 15, no. 1, p. 36259, 2025. [Online]. Available: <https://doi.org/10.1038/s41598-025-20102-w>
- [39] A. Agius Anastasi *et al.*, "Brain symmetry index in healthy and stroke patients for assessment and prognosis," *Stroke Research and Treatment*, vol. 2017, 2017. [Online]. Available: <https://doi.org/10.1155/2017/8276136>
- [40] B. Kemp, A. H. Zwinderman, B. Tuk, H. A. C. Kamphuisen, and J. J. L. Oberyé, "Analysis of a sleep-dependent neuronal feedback loop: The slow-wave microcontinuity of the EEG," *IEEE Transactions on Biomedical Engineering*, vol. 47, no. 9, pp. 1185–1194, 2000. [Online]. Available: <https://doi.org/10.1109/10.867928>
- [41] A. L. Goldberger, L. A. N. Amaral, L. Glass *et al.*, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000. [Online]. Available: <https://doi.org/10.1161/01.cir.101.23.e215>
- [42] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794. [Online]. Available: <https://doi.org/10.1145/2939672.2939785>
- [43] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems 30 (NIPS)*, 2017.
- [44] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Advances in Neural Information Processing Systems 31 (NeurIPS)*, 2018. [Online]. Available: <https://doi.org/10.48550/arXiv.1706.09516>
- [45] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.
- [46] A. Lacoste, A. Luccioni, V. Schmidt, and T. Dandres, "Quantifying the carbon emissions of machine learning," *arXiv preprint arXiv:1910.09700*, 2019. [Online]. Available: <https://doi.org/10.48550/ARXIV.1910.09700>
- [47] B. Hjorth, "EEG analysis based on time domain properties," *Electroencephalography and Clinical Neurophysiology*, vol. 29, no. 3, pp. 306–310, 1970. [Online]. Available: [https://doi.org/10.1016/0013-4694\(70\)90143-4](https://doi.org/10.1016/0013-4694(70)90143-4)
- [48] W. Chen, Y. Cai, A. Li, Y. Su, and K. Jiang, "Single-channel sleep EEG classification method based on LSTM and hidden markov model," *Brain Sciences*, vol. 14, no. 11, p. 1087, 2024. [Online]. Available: <https://doi.org/10.3390/brainsci14111087>
- [49] S. Khalighi, T. Sousa, J. M. Santos, and U. Nunes, "ISRUC-Sleep: A comprehensive public dataset for sleep researchers," *Computer Methods and Programs in Biomedicine*, vol. 124, pp. 180–192, 2016.



MD WAHIDUZZAMAN SUVA received the B.Sc. degree in computer science and engineering from American International University-Bangladesh (AIUB), Dhaka, Bangladesh, in 2025, where he is currently pursuing the M.Sc. degree in computer science. He is an Applied AI Engineer with Odin, Dhaka, where he builds LLM-based document intelligence and financial reporting systems, and a Co-Founder of Byteverse Ltd. and of Prodigy AI, a non-profit research group focused on brain-computer interfaces. He is the lead author of an article on symmetry-aware equivariant quantum clustering in *Physica Scripta* and a co-author of an article on Lorentz-equivariant quantum graph neural networks in the *IEEE Transactions on Artificial Intelligence* and of a Springer book chapter on federated learning in healthcare. His research interests include brain-computer interfaces, EEG decoding, quantum machine learning, and applied large language model systems. He was a recipient of the Rising Star Performer Award from Odin in 2025.



ESM-E MOULA CHOWDHURY ABHA received the B.Sc. degree in computer science and engineering from American International University-Bangladesh (AIUB), Dhaka, Bangladesh, in 2025, where she is currently pursuing the M.Sc. degree in computer science. She is with the Department of Computer Science and Engineering, AIUB, and a Researcher with Prodigy AI, a non-profit research group focused on brain-computer interfaces. She has over three years of industry experience as a Backend and AI Application Developer, building scalable API architectures and AI-integrated platforms for fintech, education, and customer-support products. She is the lead author of a Springer book chapter on federated learning in healthcare technology and a co-author of an article on equivariant quantum clustering in *Physica Scripta*, and she has contributed to research on attention-based EEG classification and multimodal deep learning for autonomous vehicle perception. Her research interests include machine learning for health informatics, physiological time-series analysis, quantum and federated learning, and multimodal deep learning.



MD IMTIAJ ALAM SAJIN received the B.Sc. degree in computer science and engineering, with a major in data science, from American International University-Bangladesh (AIUB), Dhaka, Bangladesh, in 2026, where he is currently pursuing the M.Sc. degree in computer science. He is with the Department of Computer Science and Engineering, AIUB, and a Researcher with Prodigy AI, a non-profit research group focused on brain-computer interfaces. He has also worked as a Software Developer with Byteverse Ltd., Dhaka, contributing to Chatsby, an AI-driven customer-support platform. His B.Sc. thesis proposed a dual-attention transformer for EEG classification in brain-computer interfaces. He is a co-author of an article on symmetry-aware equivariant quantum clustering in *Physica Scripta* and of a Springer book chapter on federated learning in healthcare technology. His research interests include deep learning, quantum machine learning, biomedical signal analysis, EEG-based brain-computer interfaces, and applied machine learning.



MD IFTEKHARUL MOBIN (Member, IEEE) is an Assistant Professor with the Faculty of Science and Technology, American International University-Bangladesh (AIUB), Dhaka, Bangladesh. He received the Ph.D. degree from Queen Mary University of London, London, U.K., and the B.Sc. degree from the Islamic University of Technology (IUT), Gazipur, Bangladesh. He was awarded the Queen Mary scholarship and the ImpactQM scholarship in acknowledgment of his contributions to industry-academia partnerships, and his research has received support from Bangladesh's Ministry of Information and Communication Technology. He served as a software engineer in the United Kingdom, Japan, and Bangladesh for several years, and that dual exposure to academic and industry settings informs his perspective on industry-academia collaboration. He has served on the program committees of international conferences and as an editor and reviewer for conferences and journals worldwide. His research interests include natural language processing, time-series prediction, health informatics, the Internet of Things, and image processing.

• • •